SoS 2026: Graph Theory

Plan of Action

P.Teenu Anand
Mentor: V.Abhinav

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Timeline

Week 1:Fundamentals of graph theory including vertices, edges, degree of a vertex, adjacency, simple graphs, complete graphs, cycle graphs, and line graphs from the provided Graph Theory notes.graph representations using adjacency lists and adjacency matrices. Implement graph input and representation in C++

Week 2 :Graph traversal methods and recursive graph exploration. Breadth First Search (BFS) and Depth First Search (DFS), connected components, and graph traversal trees. Implement BFS and DFS using adjacency lists and solve beginner graph traversal problems.

Week 3 :Bipartite graphs, matching problems, Hall's Theorem, and stable matching problems from the provided notes. Understand how graph theory is applied to real-world matching and scheduling problems.

Week 4 :Graph coloring and chromatic number. applications of graph coloring in scheduling and frequency allocation problems. Implement bipartite checking and basic graph coloring approaches.

Mid-term Report Submission

Week 5 : Paths, walks, shortest paths, and connectivity in graphs. relation between adjacency matrices and walks in graphs. Implement shortest path algorithms for unweighted graphs.

Week 6 : Weighted graphs and shortest path algorithms including Dijkstra's Algorithm. applications of weighted graphs in networks and optimization problems. Implement Dijkstra's Algorithm and solve practice problems.

Week 7 : Study trees, spanning trees, Minimum Spanning Trees (MST), and Disjoint Set Union (DSU). Learn Kruskal's Algorithm and applications of spanning trees in optimization and networks. Solve intermediate graph problems.

Week 8 :Buffer week for revision, solving additional graph problems, debugging implementations, and preparing final report. Explore selected classical graph theory problems from Wikipedia and summarize learnings

End-term Report Submission

References

1.Introduction to Graph Theory by Douglas B. West

2.Kenneth H. Rosen, Discrete Mathematics and Its Applications — Chapters on Graphs and Trees.

3.Wikipedia Graph Theory Problems Page for classical graph problems and applications.